

# iKure Health Monitoring Kiosks

**For those who are situated in rural or far-flung areas of West Bengal, access to healthcare and diagnostic services has become as easy as a visit to the local grocery shop, thanks to iKure's health kiosks**

■ JALAJA RAMANUNNI

**H**ealthcare and technology go hand in hand, but the benefits of technology-enhanced healthcare services have not yet percolated to the rural and remote areas of India. iKure Techsoft, a medical research technology company, is trying to bridge this gap.

iKure Techsoft, in partnership with Aegle Angels Foundation, is setting up intelligent health kiosks in rural areas where healthcare infrastructure is negligible. So rural masses need not travel to far away diagnostic centres for regular check-ups. So far, the company has installed 15 such kiosks in rural parts of West Bengal.

## How the kiosks work

Every kiosk has a rural medical practitioner (RMP) who takes patients' blood pressure, blood sugar, body temperature, ECG (electrocardiogram), weight, height and pulse readings with commonly used medical instruments. On the face of it, it looks like a conventional dispensary, but what makes these facilities unique is the technology that powers these kiosks.

The kiosks are equipped with many interesting and innovative pieces of technology, the most innovative being a handheld device called WHIMS—the wireless health incident monitoring system. It can capture data from a maximum of 16 medical instruments or 16 patients at the same time. The device has been configured to enable automatic digital transmission of data through Zigbee—a wireless protocol. After WHIMS reads measurements from these instruments, Zigbee auto-



WHIMS prototype demo

matically transmits all the collected information to a data server, wirelessly. This information (test reports) is stored in the server as the patient's medical history for future reference, explains Sujay Santra, founder-director, iKure Techsoft.

Based on the test results, the RMP can prescribe medicines. The RMPs are medical practitioners trained and appointed by iKure. They manually enter the analysis and prescription, which is then stored in the server. Medical history records of patients are also stored for future use.

## Enabling healthcare remotely

For complex cases, the RMP may require to consult a doctor. To deal with such situations, iKure has tied up with The Mission hospital and Jubilant Kalpataru hospital, both of which are located in West Bengal. At these hospitals, a team of about eight doctors

remotely attends to the cases referred by the RMPs.

If required, the RMP can also consult a specialist stationed at these hospitals via a phone after transmitting the patient's details. The specialist can access the patient's details directly from the server. If the patient requires special advice on a complicated case, he is referred to the nearest hospital available. In this case, the medical history of the patient along with the line of treatment prescribed by the RMP at the health kiosk (if any) is wirelessly sent to the server. This data can be accessed by the specialist at the hospital through a laptop or mobile phone.

Patients from rural areas usually do not remember details of their medical history and are unaware of medication they have taken. "Having test results and prescriptions (stored on the server) transmitted to specialists in the hospital (where the patient's case is referred)